

The Chiral Anomaly in Condensed Matter

BSc Project presentation

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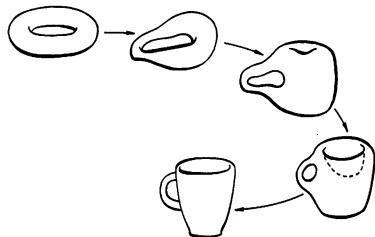
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Scope

- 1 Topological Insulators.
- 2 Dirac & Weyl semimetals.
- 3 The chiral anomaly.
- 4 Search for the chiral anomaly in CMP.

What is Topology?

- Topology is the study of reversible continuous deformations (homeomorphisms) of objects.
- Topological invariants are quantities preserved under stretching and twisting, but not necessarily tearing or stitching.



Topological insulators

- The vacuum has the simplest band structure.
- Insulators are 'conventional' or 'topologically trivial' if a homeomorphism of lattice parameters can transform the band structure into the band structure of the vacuum.
- In quantum mechanics, adiabatic processes are gradual changes in parameters which make eigenstates evolve continuously.
- Conventional insulators are adiabatically connected to the 'atomic limit'.
- Not all materials are topologically trivial!

Berry phase & Electronic structure

- The Berry phase is the change in phase picked up by the wavefunction upon completing an adiabatic loop in the Brillouin zone.

Berry phase

$$\phi = \oint_C d\boldsymbol{\lambda} \cdot i \langle u_{\mathbf{k}} | \nabla_{\mathbf{k}} | u_{\mathbf{k}} \rangle = \oint_C d\boldsymbol{\lambda} \cdot \mathbf{A}. \quad (1)$$

- In quantum mechanics phases are unphysical; only phase differences are observable, so the system is underdetermined by a Berry phase.
- This is analogous to gauge freedom in electromagnetism and the Berry potential has a gauge transformation similar to the electromagnetic four-potential.

Berry Phase & Electronic structure

Chern theorem

$$\Phi = \iint_{\Sigma} d\mathbf{S} \cdot \vec{\nabla} \times \mathbf{A} = 2\pi c; \quad c \in \mathbb{Z} \quad (2)$$

- The Chern theorem tells us that the Berry flux through a surface in the Brillouin zone is a topologically invariant integer multiple of 2π .
- Additionally, electronic band degeneracy points are monopoles of Berry flux and are thus topologically protected.

Dirac & Weyl semimetals

- Weyl semimetals (WSMs) are materials with linear dispersion, given by the Weyl Hamiltonian, at band degeneracies near the fermi energy called Weyl points.
- Weyl points have Chern numbers of $c = \pm 1$; positive and negative chirality.

Weyl point symmetries

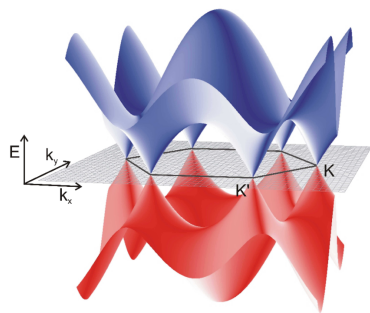
Inversion symmetry $\implies \Omega(-\mathbf{k}) = -\Omega(\mathbf{k})$.

Time-reversal symmetry $\implies \Omega(-\mathbf{k}) = \Omega(\mathbf{k})$.

- Dirac semimetals (DSMs) are WSMs which have both inversion and time-reversal symmetry leading to band crossings being four overlapping Weyl points (Dirac points).

Dirac & Weyl semimetals

- The most famous 2D DSM is graphene which has 6 Dirac points at the edge of the Brillouin zone.
- The Hamiltonian is the massless Dirac Hamiltonian; thus, quasiparticles in graphene are observed as massless Dirac fermions [1].
- This suggests WSM quasiparticles should exhibit the effects of relativistic Weyl fermions. In particular, the chiral anomaly.



[2]

What is an anomaly?

- An anomaly is a symmetry of a classical theory which is broken in the associated quantum field theory.
- Noether's theorem tells us that symmetries correspond to physical conservation laws.
- Thus, an anomaly can be seen as the existence of quantum processes which violate a classical conservation law.

The Chiral Anomaly

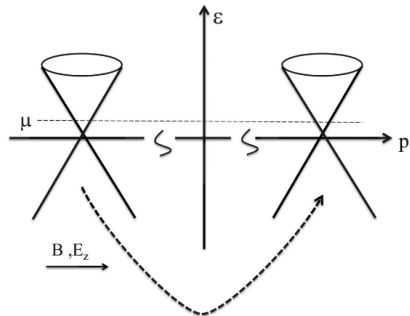
- The Dirac equation has two symmetries which correspond to two conserved currents; the axial and vector currents (j & j_5).
- The vector current is the sum of the currents of opposite chirality whereas the axial current is the difference.
- The axial symmetry is anomalous; the axial current is not conserved.

Chiral anomaly

$$\vec{\nabla} \cdot j_5 = \frac{e^2}{4\pi^2\hbar^2} \mathbf{E} \cdot \mathbf{B} \quad (3)$$

The Chiral Anomaly

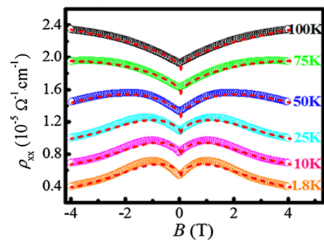
- In high energy physics, the chiral anomaly manifests as the fast π^0 decay rate called the Adler–Bell–Jackiw anomaly [3, 4].
- In WSMs the chiral anomaly is observed as a large negative magnetoresistance as the external fields cause electrons to flow between Weyl points of opposite chirality.



[5]

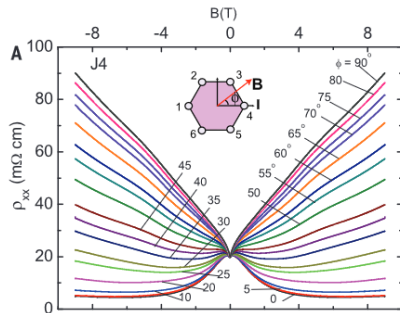
Experiments

- TaAs was the first WSM discovered and exhibits a weak negative linear magnetoresistance (LMR) [6].
- However, the high carrier mobility of TaAs and other WSMs produces current jetting which also create negative LMRs [7].



Experiments

- The Na_3Bi is a DSM however, an external $\mathbf{B}$ field breaks the time-reversal symmetry splitting the two Dirac points into Weyl points[8].
- Moreover, the low carrier mobility excludes current jetting effects making the negative LMR in Na_3Bi the strongest evidence for the chiral anomaly in condensed matter.



References

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